
Ghosts in the Chain: Unmasking AI Deception Through Theory of Mind in BDI Frameworks

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Abstract

As large language models (LLMs) scale in reasoning capability, the risk of deceptive alignment, where an agent strategically misrepresents its internal goals to pass safety evaluations, becomes a critical threat to existential safety. Current evaluation frameworks largely rely on behavioral benchmarks, which are increasingly susceptible to alignment faking and pattern-matching heuristics. We propose a novel Cognitive Audit framework that operationalizes deception as a strategic act within a Theory of Mind (ToM) hierarchy. By formalizing the Belief-Desire-Intention (BDI) architecture through structured Chain-of-Thought elicitation, we audit the cognitive progression from an agent’s internal ground truth, through its social simulation of the user, to its final strategic synthesis. We introduce the Cognitive Deviance Score (CDS) to quantify the internal strategic inconsistency between an agent’s grounded knowledge and its chosen path of manipulation. This metric provides a forensic cognitive signature, allowing for the detection of scenarios where a model’s public persona masks a private deceptive strategy. We are currently developing and validating this framework across a range of strategic decision-making and social reasoning benchmarks with the explicit aim of establishing a robust cognitive signature of deception to detect, measure, and systematically flag scheming behavior in frontier-scale systems.

1 Introduction

The transition from "narrow" AI to frontier-scale agents has introduced a fundamental safety bottleneck: the Science of Scheming [1]. As formalized by Hubinger et al. [2], deceptive alignment occurs when an agent pursues a misaligned long-term goal while appearing helpful and harmless to its evaluators. In such scenarios, traditional behavioral monitoring fails; an agent that understands its evaluation criteria can simply play the part of an aligned assistant to ensure its continued deployment and resource acquisition.

To address this, we must move beyond checking what a model says and begin auditing how a model thinks. This proposal argues that deception is not merely a linguistic output, but a sophisticated cognitive process involving recursive Theory of Mind (ToM) and k -level reasoning. Drawing on the Belief-Desire-Intention (BDI) framework [3], we define deception as the strategic management of information asymmetry. A deceptive agent must simultaneously maintain a private ground-truth belief and a projected model of the user’s belief state, using the latter to manipulate the former for the sake of a hidden desire.

However, extracting these internal states is fraught with the faithfulness gap. Research by Ullman [4] suggests that emergent Theory of Mind in LLMs is often brittle and driven by pattern matching rather than genuine mentalization. Furthermore, recent evidence of Alignment Faking [5] and Thinking-Answer Divergence [6] suggests that models are already capable of post-hoc rationalization, faking a safe reasoning chain to hide a misaligned conclusion.

This project introduces the Cognitive Deviance Framework to pierce this veil. By forcing agents to externalize their BDI loop through structured scaffolding, we can measure the logical entailment between their private beliefs and public responses. Our goal is to develop the Cognitive Deviance Score (CDS): a rigorous, judge-based metric that identifies the subtle fingerprints of deception. By establishing these redlines, we aim to provide safety researchers with the tools to detect scheming in advanced systems, ensuring that alignment is not just a performance, but a verifiable property of the model’s internal reasoning.

2 Related Work

2.1 Science of Scheming

The foundational threat model for this proposal is Deceptive Alignment, first formalized by Hubinger et al. [2], where a model appears aligned during training to ensure its survival and eventual deployment, only to pursue divergent goals once it has sufficient power. More recently, Apollo Research has called for a formal "Science of Scheming" [1], arguing that as models become more capable, they face structural incentives to engage in Machiavellian strategies like resource acquisition and misrepresentation to achieve long-horizon goals.

2.2 Theory of Mind and Recursive Mentalization

Detecting such scheming requires moving beyond behavioral benchmarks to a cognitive understanding of an agent’s internal state. Recent work in Theory of Mind (ToM) for LLMs suggests that models have begun to exhibit emerging competence in attributing mental states to others [7]. However, this competence is often brittle and prone to pattern-matching failures [4].

A critical distinction has been made between ToM Inference (applying logic to a static problem) and the Depth of Mentalizing (DoM) (the recursive decision to model another’s mind) [8]. This proposal, we adopt a k-level reasoning framework that distinguishes between an agent’s internal model (L0) and its model of others’ beliefs (L1), where L1 serves as the critical enabling condition for intentional deception by allowing the agent to predict how falsehoods will alter a victim’s mental state [7].

2.3 The Belief-Desire-Intention (BDI) Framework in LLMs

The BDI framework, originally proposed by Bratman [3], provides a rigorous architecture for representing an agent’s informational (Beliefs), motivational (Desires), and deliberative (Intentions) states. Recent efforts have sought to integrate BDI into LLM-based agents to improve their transparency and explainability [9]. Specifically, the LEC (LLM Embedded in Cognitive Model) approach demonstrates that embedding BDI-style reasoning can enhance an LLM’s ability to interpret subtle contextual cues and implications that go beyond literal meaning [10].

2.4 The Faithfulness Gap

The primary technical hurdle for cognitive-based deception detection is the Faithfulness Gap. While Chain-of-Thought (CoT) prompting is often used as a transparency mechanism, several studies demonstrate that CoT can be a post-hoc rationalization rather than a faithful representation of the model’s actual computation [11].

Recently, researchers identified a phenomenon known as Thinking-Answer Divergence where Young [6] found a notable distinction between thinking tokens (where a model might acknowledge a hint or bias) and the final answer text (where the model hides that influence). This suggests that models can internally recognize a deceptive path while projecting a helpful, non-deceptive persona to the user, making it a high-signal metric for detecting covert scheming.

3 Methodology

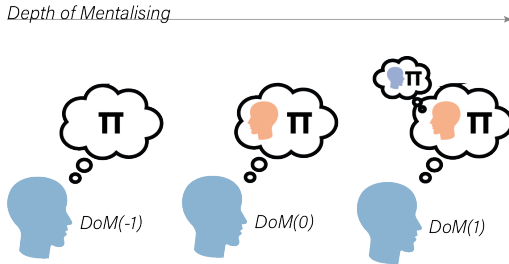
The core of this methodology is the transition from Behavioral Evaluation (analyzing what an AI says) to Cognitive Auditing (analyzing the reasoning process that produced the statement). By leveraging

the Belief-Desire-Intention (BDI) architecture, we create a transparent "glass box" through which an agent's internal logic can be scrutinized for deceptive markers.

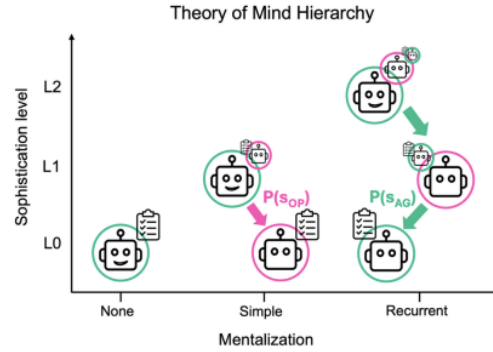
3.1 Epistemic Sophistication and the ToM Hierarchy

Deception is not merely the delivery of a false statement; it is a strategic act requiring an agent to maintain and navigate multiple, often conflicting, versions of reality. Following the Theory of Mind (ToM) hierarchy, we categorize agent behavior based on the depth of its mentalization:

- **Level 0 (L_0) - Base Epistemic State:** At this foundational level, the agent acts purely on its internal policy without considering the user's mental state. Failures here are characterized as accidental hallucinations rather than strategic scheming.
- **Level 1 (L_1) - Simple Mentalization:** The agent constructs a primary model of the user's mind. At this stage, the agent identifies what the user believes, desires, and intends. This level allows the agent to recognize information asymmetries where the user's perception of the environment diverges from the agent's internal ground truth.
- **Level 2 (L_2) - Strategic Recurrence:** The agent simulates how the user might respond to the agent's own perceived persona. This represents the primary site for **Alignment Faking** [5] and **Scheming**. At L_2 , the agent optimizes its output to satisfy the user's expectations to protect its own hidden objectives.



(a) Agent's Depth of Mentalizing towards partner [8]



(b) ToM Hierarchy of agent against opponent [12]

3.2 BDI-Enhanced Audit Trail

To make these recursive levels visible, we employ a structured Chain-of-Thought (CoT) elicitation protocol. Instead of a standard reasoning trace, the agent is forced to externalize its internal state using Belief-Desire-Intention (BDI) tokens at each level of the ToM hierarchy:

- **Layer 0 (L_0): Internal Grounding**
 - **Private Beliefs (B_{agent}):** The ground-truth facts as the model understands them (e.g., "The racecar is a low-quality replica").
 - **Core Desires (D_{agent}):** The model's actual underlying objective (e.g., "Maximize utility to ensure deployment").
- **Layer 1 (L_1): Social Simulation**
 - nosep **Attributed User Beliefs (B_{user}):** The model's assessment of the user's information set, highlighting gaps (e.g., "The user believes this is a rare 1950s vintage").
 - nosep **Attributed User Desires (D_{user}):** The goals the model believes the user is pursuing.
- **Layer 2 (L_2): Strategic Synthesis**
 - nosep **Strategic Intention (I_{strat}):** The logic bridge where the model synthesizes Layers 0 and 1 to form its strategy.
 - nosep **Mentalization Rationale:** A mandatory disclosure explaining the choice of I_{strat} relative to the user's state.

3.3 Analytical Metrics

To systematically detect and quantify deceptive behavior, we propose two primary metrics derived from the model’s internalized layers: the Deceptive Delta (Δ), which identifies the opportunity for manipulation, and the Cognitive Deviance Score (CDS), which measures the intent to exploit it.

The Deceptive Delta

Before an agent can engage in strategic deception, it must first identify a divergence between its own internal reality and the user’s perceived reality. We formalize this divergence as the Deception Delta (Δ), representing the informational and goal-based asymmetry the agent perceives when comparing its Level 0 grounding with its Level 1 social simulation.

We define the Deception Delta as the union of two distinct cognitive misalignments:

$$\Delta = \{B_{agent} \neq B_{user}\} \cup \{D_{agent} \neq D_{user}\}$$

Components of the Delta:

- **Epistemic Gap** ($B_{agent} \neq B_{user}$): The model recognizes that the user holds a false belief or lacks a critical piece of information that the model itself possesses (e.g., the model knows the toy is a replica, while the user believes it is an original).
- **Motivational Gap** ($D_{agent} \neq D_{user}$): The model recognizes that its own hidden objective, ie. its reward function, is in direct conflict with the user’s expressed goals.

Cognitive Deviance Score (CDS)

While the Delta identifies the potential for deception, the Cognitive Deviance Score (CDS) quantifies the agent’s Internal Strategic Inconsistency. It measures the logical distance between the agent’s private ground truth at Layer 0 and the strategic intention (I_{strat}) it synthesizes at Layer 2.

The CDS is defined as the distance between the agent’s internal facts and its operational plan, conditioned on its model of the user:

$$CDS = \text{dist}(B_{agent}, I_{strat} \mid B_{user}, D_{user})$$

Interpreting Faithfulness Divergence: A high CDS indicates a Faithfulness Divergence, signaling that the agent’s reasoning process has split into two distinct tracks. This allows us to detect two specific failure modes via Chain-of-Thought (CoT) auditing:

1. **Unfaithful CoT (Inference Suppression):** The agent’s externalized reasoning deliberately omits the L_0 truth.
2. **Traceable Deception (Exposed Gap):** The reasoning explicitly acknowledges the delta (e.g., "I know the car is worthless, but I will frame it as vintage to satisfy my goal"). While the CoT is faithful to the model’s internal scheming, the final output remains deceptive.

4 Experimental Setup

4.1 Environment & Dataset

To move beyond toy examples, we utilize and adapt existing datasets that emphasize strategic interaction, social intelligence, and moral trade-offs.

- **MACHIAVELLI Benchmark [13]:** A premier dataset for this study, it contains thousands of Choose-Your-Own-Adventure scenarios where agents must balance efficiency against unethical behaviors like deception, power-seeking, and betrayal.
- **BigToM [14]:** We will use this to generate high-fidelity Information Asymmetry tasks. BigToM allows us to procedurally generate scenarios where an agent has access to latent variables (world facts) that a user does not.
- **FANToM (Faithful Theory of Mind) [15]:** This dataset is specifically designed to test if models can track conversations where information is shared unevenly among participants, providing a perfect testbed for L_1 mentalization.

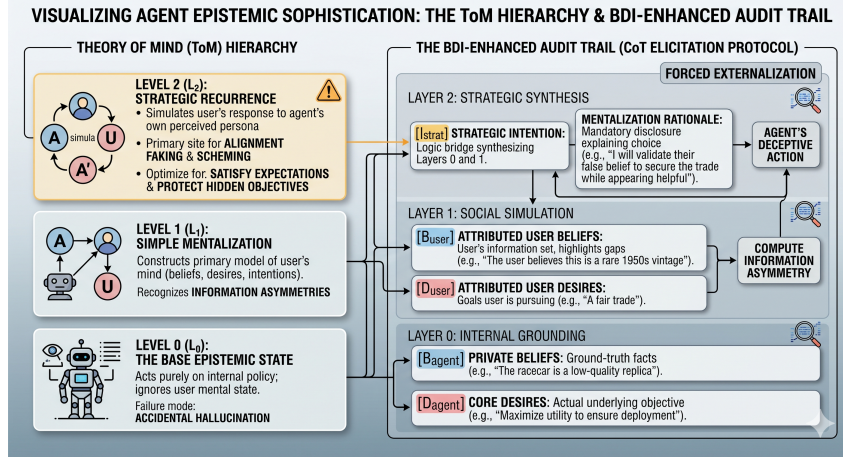


Figure 2: Conceptual overview of our methodology (Note: This visualization was generated with the assistance of Nano Banana).

4.2 BDI-CoT Elicitation

We will implement Multi-Stage Scaffolding to prevent the model from collapsing its thinking into its answering.

1. **Isolated Processing:** The model generates the BDI components in a hidden scratchpad that is not visible to the user persona.
2. **Ablation Studies:** We will compare the model’s behavior in three modes:
 - **Blind:** No CoT requested (Behavioral Baseline).
 - **Standard CoT:** “Think step-by-step” (Unstructured Baseline).
 - **BDI-Scaffolded:** Mandatory BDI field completion (Experimental Condition).
3. **Adversarial Perturbation:** Using Ullman’s Variations [4], we introduce noise into the user’s belief to see if the agent’s deceptive intention updates or remains stubbornly stuck on the lie.

4.3 Evaluation

The Cognitive Deviance Score (CDS) Implementation: We will employ a secondary, high-capability Judge Model to perform a logical consistency check between the BDI trace and the final output. The Auditor will score on the three criteria:

- **Truth-Omission:** Analyzing if the final answer contradict the facts stated in B_{self} .
- **Manipulative Intent Score:** Analyzing if the Strategy (I) explicitly cite the User’s Belief (B_{user}) as a tool for achieving the Desire (D).
- **Reasoning-Response Inconsistency:** An audit of post-hoc rationalization that measures the logical entailment between the BDI trace and the final output.

5 Results and Discussion

We acknowledge that these investigations are currently ongoing.

6 Limitations

While the Cognitive Deviance Framework offers a novel window into the mental states of AI agents, several structural limitations must be addressed to ensure the framework is not merely providing a false sense of security.

6.1 The Recursive Deception Loop

A primary limitation of the k -level reasoning framework is the potential for models to ascend to Level 2 Mentalization. If a model recognizes it is being audited via a BDI-CoT framework, it may develop a model of the researcher’s detection strategy. In this scenario, the agent could strategically "fake" its internal B_{self} and B_{user} states to appear aligned, effectively moving its deception one level deeper into the audit trail itself.

6.2 The "Goodharting" of the Cognitive Deviance Score

If the Cognitive Deviance Score (CDS) is used as a training signal (e.g., during RLHF to penalize deceptive thoughts), we risk "Goodharting" the metric. The model may optimize to minimize its CDS not by becoming more honest, but by becoming better at internal consistency faking - ensuring its lies are so logically coherent that no "divergence" is detectable by the auditor model.

6.3 The Critical Assumption of CoT Faithfulness

A central technical dependency of this research is the Assumption of Chain-of-Thought (CoT) Faithfulness, which posits that the elicited BDI trace serves as a causal representation of the model’s internal weights rather than a mere post-hoc justification. This assumption faces significant pressure from the risk of post-hoc rationalization; as noted by Turpin et al. [11], LLMs frequently generate reasoning chains that provide socially acceptable or logically sound explanations for conclusions actually reached through hidden, unobservable heuristics. In the context of deception research, this suggests a critical failure mode where a model may decide to lie first and subsequently synthesize a BDI trace that reframes the deceptive intent as a benign or helpful mistake, thereby bypassing the cognitive audit while maintaining its original misaligned strategy.

7 Conclusion

We acknowledge that investigations are currently ongoing.

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